COMPUTER ENGINEERING DEPARTMENT

Subject : DS **ASSIGNMENT NO – 03** **Unit : III**

1. What is **stack** ? Write an **ADT for stack**.(Pseudo code of basic stack operation)
2. What is **recursion** ? Explain use of stack for recursion, types of recursion with example.
3. Write pseudo-Python/**C/C++ code to implement stack using array** with overflow and underflow conditions.
4. What is **backtracking algorithm design strategy**? How stack is useful in Backtracking. Explain use of backtracking in 4-Queen's problem.
5. Convert the **following prefix into postfix**

   *+a – bc / - de + - fgh
6. What is **infix, prefix and postfix expression**? Give example of each. Explain **evaluation of postfix expression** with suitable example expression and assume values for variables used to solve it.

7. Explain **the stepwise conversion using stack for the given infix expression to the postfix expression :**

   1) ((a/(b-c+d))*(e-a)*c

   2) (a+b) * d + e/(f + a*d) + c.

8. Define **Queue(Simple queue).** Explain Queue as an ADT. Write pseudo Python/C/C++ code to represent queue operations.
9. Draw and explain implementation of **Linear Queue** using Singly Linked List. Explain Add, Remove, Queue Full and Queue Empty operations
10. What are advantages of **Circular Queue over Linear Queue** using static memory allocation? Write pseudo code to add and remove element from Circular Queue along with Queue Full and Empty condition

11. What is **Doubly Ended Queue**? Draw Diagram with labelling four basic operations at appropriate places. Which two data structures are combined in it and how?
12. What is **Priority queue**. Explain Array implementation of priority queue with all basic operations(ADT). State its applications.
13. Consider the following queue where queue is circular queue having 6 memory cells

    Front = 2 , Rear = 4

    Queue : -- , A , C , D , -- , --

    Describe the queue as following operation take place :

    1) F is added

    2) Two letters are deleted

    3) R is added to the queue

    4) S is added to the queue

    5) One letter is deleted.

14. Compare **Singly, Doubly and Circular** Linked List.
15. Write pseudo Python/C/ C++ code to represent **singly linked list** as an ADT.
16. What is **doubly linked list**? Explain the process of deletion of an element from doubly linked list with example(Pseudo code in Python/C / C++).
17. What is **Dynamic Data Structure**? Explain the **Circular Linked list** with its operation.(Pseudo code in Python/C / C++)

********** **Best of Luck** **********